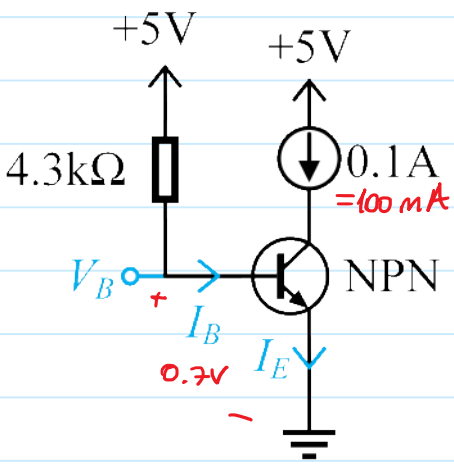
Q1 15pts



ACTIVE

$\beta = 100$

$V_B = ?$

$I_B = ?$

$I_E = ?$

active mode $\Rightarrow$ $V_{BE} = \boxed{V_B = 0.7V}$ 5 pts

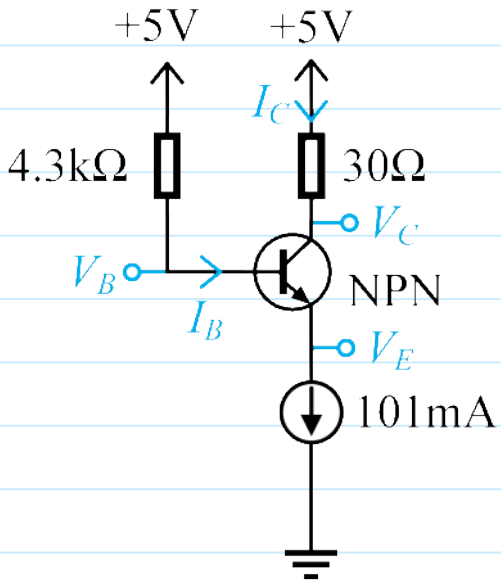
$I_B = (5 - 0.7)/4.3k = \boxed{1mA = I_B}$ 5pts

OR active mode $\Rightarrow$ $I_C = \beta . I_B \Rightarrow I_B = I_C/\beta = \frac{100mA}{100} = 1mA$

$I_B + I_C = 100mA + 1mA = \boxed{101mA = I_E}$ 5pts

OR $I_E = (\beta + 1) I_B = (101) . 1mA = 101mA = 0.101A$

Q2 30 pts



ACTIVE

$\beta = 100$
$I_B = ?$
$V_B = ?$
$V_E = ?$
$I_C = ?$
$V_C = ?$

active mode ⇓

$I_B = I_E/(\beta+1) = 101\,mA/(100+1) = 1mA = I_B$

5pts 5pts

$V_B = 5 - 4.3k \cdot I_B = 5 - 4.3k \cdot 1mA = 5 - 4.3 = 0.7V = V_B$

5pts

$V_{BE} = V_B - V_E = 0.7V = 0.7 - V_E \Rightarrow V_E = 0V$

5pts

active mode ⇓

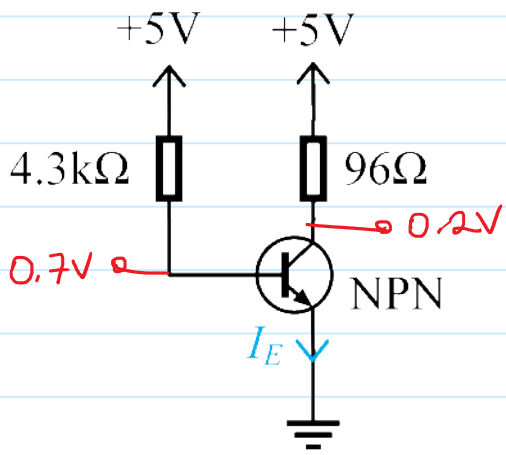
$I_C = \beta \cdot I_B = 100 \cdot 1mA = 100mA = I_C$

5pts

$V_C = 5 - 30 \cdot I_C = 5 - 30 \cdot (0.1) = 5 - 3 = 2V = V_C$

5pts

Q3 5pts



SATURATION
$V_{CE} = 0.2V$

$I_E = ?$

1pts

$V_{CE} = 0.2V = V_C - V_E = V_C - 0 \Rightarrow V_C = 0.2V$

$I_C = (5 - 0.2)/96 = 0.05A = 50mA$ 1pts

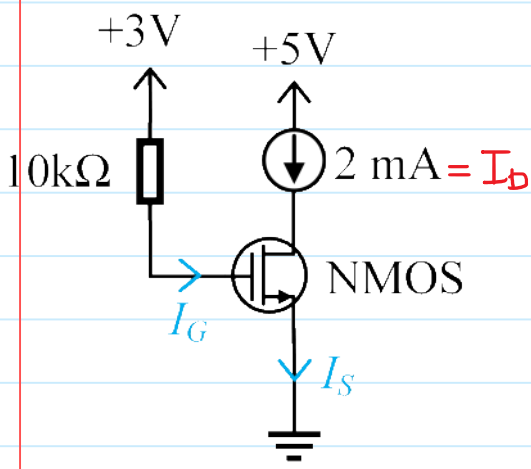
$V_{BE} = V_B - V_E = V_B - 0 = 0.7 \Rightarrow V_B = 0.7V$ 1pts

$I_B = (5 - 0.7)/4.3k = 1mA$ 1pts

$I_E = I_B + I_C = 1mA + 50mA = \boxed{51mA = I_E}$

1pts

Q4 15 pts



SATURATION

$I_D = \frac{1}{2} k_n V_{OV}^2$

$V_{OV} > 0$

$V_T = 1$ V

$k_n = 1$ mA/V^2

$V_{OV} = ?$

$I_G = ?$

$I_S = ?$

3pts

$I_D = \frac{1}{2}(1\,mA/V^2) \cdot (V_{ov}^2) = 2mA$

$V_{ov} = \pm 2V$ & $V_{ov} > 0 \Rightarrow$ **$V_{ov} = 2V$**

2 pts

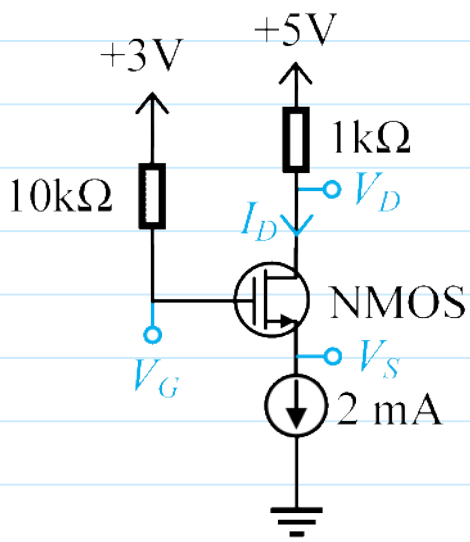
$I_G = 0A$ for MOS transistors

5pts

$I_S = I_G + I_D = 0 + I_D =$ **$2mA = I_S$**

3pts 2pts

Q5 30 pts



SATURATION

$I_D = \frac{1}{2} k_n V_{OV}^2$

$V_{OV} > 0$

$V_T = 1$ V

$k_n = 1$ mA/V^2

$V_G = ?$

$I_D = ?$

$V_D = ?$

$V_{OV} = ?$

$V_S = ?$

$I_G = 0 \Rightarrow V_G = 3 - 10k\Omega \cdot I_G = 3 - 10k \cdot 0 = 3V = V_G$

1 pts — 2 pts — 2 pts

$I_D = I_S + I_G = I_S + 0 = I_S \Rightarrow \boxed{I_D = 2mA}$ 2 pts

3 pts

$V_D = 5 - I_D \cdot 1k\Omega = 5 - 2mA \cdot 1k\Omega = 5 - 2 = \boxed{3V = V_D}$

3 pts — 2 pts

$I_D = \frac{1}{2} \cdot (1mA/V^2) \cdot V_{ov}^2 = 2mA \Rightarrow V_{ov}^2 = 4$

2 pts

$V_{ov} = \pm 2V$ & $V_{ov} > 0 \Rightarrow \boxed{V_{ov} = 2V}$

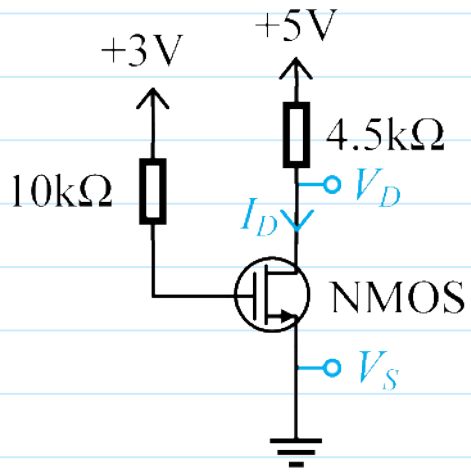
1 pts — 2 pts

$V_{ov} = V_{GS} - V_T = V_G - V_S - V_T = 2 = 3 - V_S - 1$ (V_G, V_T)

3 pts

$V_{ov} = 2 = 2 - V_S \Rightarrow \boxed{V_S = 0\,V}$ 2 pts

Q6 5 pts



LINEAR

$I_D = k_n V_{OV} V_{DS}$

$V_{OV} > V_{DS} > 0$

$V_T = 1\ V$

$k_n = 1\ mA/V^2$

$V_{DS} = 0.5V$

$I_D = ?$

$V_{DS} = 0.5V \Rightarrow V_D = 0.5V$ 2pts

$I_D = \frac{5 - 0.5}{4.5k} = \boxed{1mA = I_D}$ 3pts

OR

2 pts

$I_D = k_n . V_{OV} . V_{DS} = (1mA/V^2)(V_{GS} - V_T) . V_{DS}$

$V_D - 0$

3 − 1

$I_D = (2m) V_{DS} = (2m)(0.5) = \boxed{1mA = I_D}$ 3pts

$I_D = (2m) V_D = (5 - V_D) / 4.5k$

$9V_D = 5 - V_D \Rightarrow 10V_D = 5 \Rightarrow V_D = 0.5V$ ✓